Supplementary Information
When crop rankings do not determine land allocations

Author names and affiliations to be confirmed before submission

Stage II final scientific draft

1 Canonical theory and proof record

This section records the assumptions and arguments used in the main text. The versioned line-by-line proof package is in `theory/repaired/proofs.md`; theorem identifiers and transitions from the teacher Draft are in the canonical theorem registry. Historical numerical thresholds are not premises.

1.1 Objects and assumptions

Let $X \subset \mathbb{R}^n$ be the operational set in Eq. (2) of the main text. We assume X is non-empty and compact, $\widetilde{\pi}_i$ is integrable for every crop, and at least one $x \in X$ satisfies $r_\alpha(x) \leq \kappa$. A suitability vector s supplies only a weak order. It is not assumed to equal expected profit, a utility index or a Lagrangian derivative. The risk functional is loss-CVaR, so larger values are worse.

For a pair with $s_i > s_j$, define $h_{ij}(x) = x_j - x_i$ and tolerance $\tau_x \geq 0$. The possible and universal reversal indicators are generated by

$$g_{ij}^{+} = \max_{x \in S_\kappa} h_{ij}(x), \qquad g_{ij}^{-} = \min_{x \in S_\kappa} h_{ij}(x). \tag{1}$$

Possible reversal is $g_{ij}^{+} > \tau_x$; universal reversal is $g_{ij}^{-} > \tau_x$. A deterministic selector $\sigma(S_\kappa)$ defines selected reversal. This construction remains valid when the optimal set is a face rather than a point.

Theorem 1 (Existence and finite-scenario linearity)**.** *Under the assumptions above, the loss-CVaR problem has an optimizer and a convex feasible set. Under a finite distribution with positive scenario weights, it has an auxiliary linear-programme representation.*

Proof. Loss-CVaR is finite and continuous on the compact operational set under integrability. Its sublevel set intersected with X is closed, convex and, by assumption, non-empty. Compactness and continuity of the linear objective give existence. For scenarios $s = 1, \ldots, m$, substitute

$$r_\alpha(x) = \min_{v,q} \left\{ v + \frac{1}{1-\alpha} \sum_s w_s q_s : q_s \geq L_s(x) - v,\ q_s \geq 0 \right\}, \tag{2}$$

where v is free. The resulting objective and constraints are linear [Rockafellar and Uryasev, 2000, 2002]. □

Proposition 1 (Ordinal insufficiency)**.** *A weak or strict suitability ranking alone does not identify a cardinal acreage allocation.*

Proof. The ranking retains only pairwise order relations. A strictly increasing transformation preserves all relations while changing cardinal score ratios. More directly, choose two non-empty compact feasible sets or two linear objectives that share the same external ranking but have different unique maximizers. The ranking is compatible with both allocations. Therefore a mapping, objective, feasible set and selection rule are required. □

Proposition 2 (Feasibility-forced reversal)**.** *For $s_i > s_j$, if $\sup_{x\in X} x_i + \tau_x < \inf_{x\in X} x_j$, every feasible allocation universally reverses the pair.*

Proof. The premise implies $x_i + \tau_x < x_j$ for all $x \in X$. Since $S_\kappa \subseteq X$, the inequality holds at every optimum. The simpler bound condition $u_i + \tau_x < \ell_j$ implies the premise. □

Proposition 3 (Slack-risk intersection)**.** *Let $S_0 = \arg\max_{x\in X} \mu^\top x$. If S_0 contains a risk-feasible point, the risk-constrained optimal value equals the unconstrained value and $S_\kappa = S_0 \cap \{x : r_\alpha(x) \le \kappa\}$.*

Proof. A feasible member of S_0 attains the unconstrained upper bound in the constrained problem. Hence the values agree. Every constrained optimizer must attain that common value and therefore belongs to S_0; the reverse inclusion holds for risk-feasible members. One slack selected optimum does not imply that every member of S_0 is risk-feasible. □

Theorem 2 (Subgradient KKT characterization)**.** *Under a convex constraint qualification, or for a feasible bounded finite-scenario linear programme, a feasible x^* is optimal if and only if non-negative multipliers $\lambda, \gamma, \beta, \eta, a, b$ and $d \in \partial r_\alpha(x^*)$ satisfy stationarity and complementary slackness as stated in the main text.*

Proof. Minimize $-\mu^\top x$. The normal cone of the lower and upper bounds contributes $-a + b$; shared restrictions contribute $G^\top \eta$; land and budget contribute $\gamma\mathbf{1}$ and βc. Convex KKT conditions are necessary and sufficient. Pairwise subtraction cancels only γ. Under finite scenarios a valid CVaR subgradient is the negative scenario-profit average under tail weights whose total is one, whose upper bounds are $w_s/(1-\alpha)$ and whose atom weights may be fractional. □

Theorem 3 (Restricted dependence ordering)**.** *Fix marginals and a named copula family. If $L_{\theta_1}(x) \preceq_{\mathrm{cx}} L_{\theta_2}(x)$ for every allocation in a declared domain, loss-CVaR is weakly ordered there. If the domain is all of X, the more dispersed model has a subset of the risk-feasible policies and weakly lower optimal expected profit.*

Proof. CVaR is monotone under convex order for integrable losses. Pointwise risk ordering yields nested risk-feasible sets; maximizing the same linear objective over nested sets yields the value inequality. Crop-specific acreage need not be ordered because optimizers can move along different faces. □

Proposition 4 (Crossing-set inclusion)**.** *For any declared selector, the universal reversal region is contained in the selected region, which is contained in the possible region.*

Proof. The selected optimizer is an element of the optimal set. If the minimum gap exceeds tolerance, every element, including the selection, reverses. If the selected element reverses, the maximum gap exceeds tolerance. The argument implies no connectedness, monotonicity or uniqueness in the parameter indexing the sets. □

Theorem 4 (Information actionability)**.** *When ignoring a signal remains feasible, value of information is non-negative. A common posterior-optimal action gives zero value. Informed and uninformed optimized values are weakly non-decreasing under nested action sets, but their difference need not be monotone or strictly positive.*

Proof. The uninformed optimizer is an admissible constant contingent policy, so posterior optimization weakly improves the outer expectation. A common posterior optimizer makes the objectives equal. Set inclusion weakly raises each maximum separately, but the difference of two weakly increasing functions has no general monotonicity. E6 supplies constructive positive, zero and negative interactions under this theorem. □

1.2 Exact mechanism constructions

The four cases in main Fig. 2 use normalized two-crop acreage $x_1 + x_2 = 1$. In the margin construction, an external ranking $s_1 > s_2$ coexists with $\mu_2 > \mu_1$, so the profit optimum lies in $x_2 > x_1$. In the operational construction, bounds place all feasible points beyond the diagonal. In the risk construction, an admissible loss-CVaR half-space removes the unconstrained vertex and exposes a reversing boundary point. In the set-valued construction, the objective is constant along a feasible face crossing the diagonal. That face contains both reversal statuses, separating possible from universal reversal and making the selected status rule-dependent.

1.3 Teacher-Draft transitions

The original ranking-reversal equivalence is replaced by the KKT identity plus explicit displacement and feasibility certificates. Scalar tail monotonicity is replaced by restricted convex ordering. A unique threshold is replaced by possible and universal crossing sets. Pseudo-diversification remains a preregistered diagnostic without welfare content. Strict information–flexibility complementarity becomes a conditional hypothesis; the general theorem supports only non-negative information value when ignoring the signal is feasible.

2 Confirmatory simulation protocol and complete evidence ledger

2.1 Design, sequential rule and estimands

The Stage II design identifier and its SHA-256 hash are frozen in `simulation/configs/stage_ii_confirmatory.yaml`. Experiments E1–E6 each define their factors, contrasts, binary or continuous estimands, family-wise coverage, multiplicity, sequential check points, precision targets and maximum replication count. The execution writes 2320 raw result rows, 5360 contrast rows, 1632 scenario-registry rows, optimal-face audits, frontier records, dependence diagnostics, information cells and attribution coalitions.

At each registered check, continuous intervals use the frozen family-wise t critical value and binary intervals use the registered family-wise rule. An experiment stops for precision only when all primary intervals pass. Otherwise it proceeds to the next checkpoint until the maximum is reached. The evidence status is assigned mechanically from the complete experiment, not from selected contrasts. E2 and E6 stop at $n = 16$ with precision achieved. E1, E3, E4 and E5 reach $n = 64$ and fail.

2.2 E2 operational intervention family

E2 holds the profit law, external corn-over-soybean ranking and deterministic selection rule fixed. The base cell has no budget, rotation, contract or corn-cap intervention. Factorial cells switch budget, rotation and contract indicators; a separate corn-bound anchor supplies an explicit bound mechanism. The expected-profit base selects corn share 1. Contract or rotation displacement caps corn at 0.45 and requires soybean share 0.55. The tight budget selects corn 0.318 and soybeans 0.682. Winter wheat remains zero in these constructed cells.

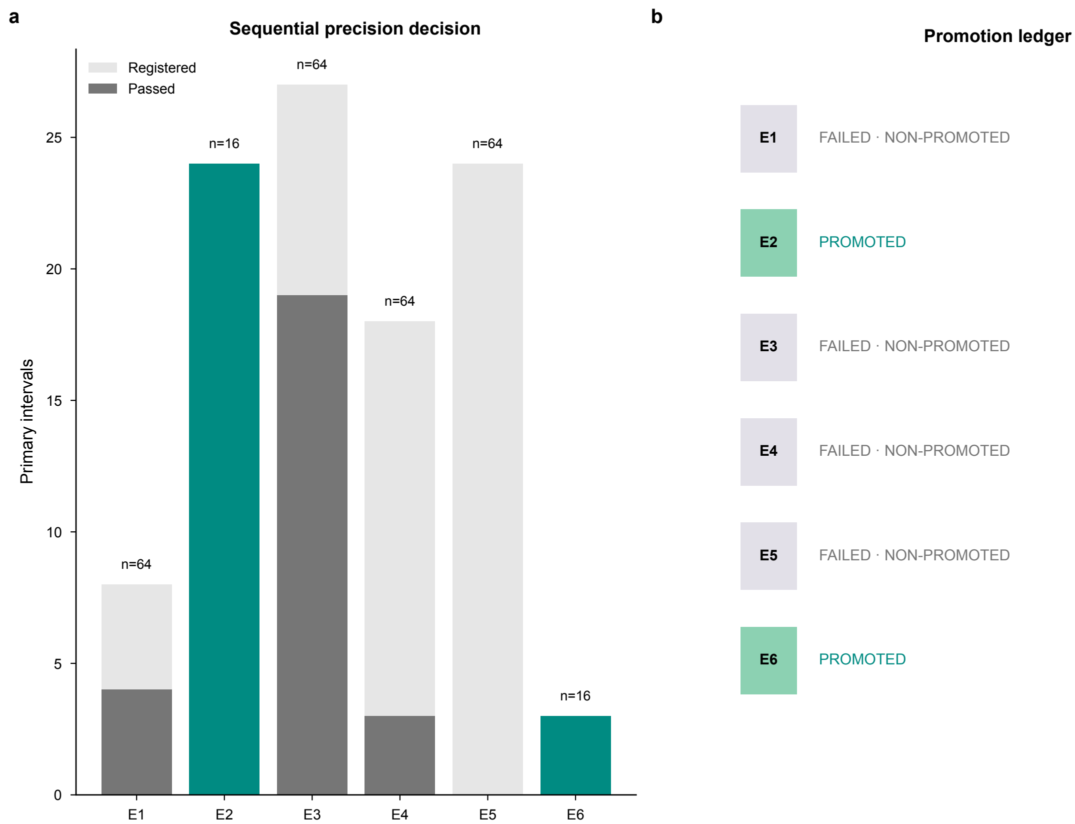

Figure 1: **Complete precision-gate ledger.** Final sequential outcomes for E1–E6. Only E2 and E6 pass every registered primary interval; failed experiments reach $n = 64$ and remain non-promoted.

Each of the eight factorial contrasts reports allocation L_1 change, expected-profit change and selected-reversal change, giving 24 registered primary intervals. Every interval passes. The selected-reversal change is one with a multiplicity-adjusted lower bound of 0.628; the continuous allocation differences are deterministic conditional on the registered cell. The optimal-face audit verifies possible, selected and universal reversal together. The mechanism labels distinguish direct forcing, marginal pressure and inactive-in-cell restrictions; a named restriction can be present without being the active marginal term.

2.3 E6 information–flexibility family

E6 computes four optimized values in each constructive environment: uninformative/restricted, informative/restricted, uninformative/flexible and informative/flexible. The interaction is

$$\Delta_{I\times F} = [V(I,F) - V(U,F)] - [V(I,R) - V(U,R)]. \tag{3}$$

The specialization-unlocks construction gives 3.258 [2.788, 3.728]. The robust-option construction gives -3.114 [-3.476, -2.751]. The dominated-option construction gives exactly 0.000. All three family-wise intervals pass at $n = 16$. Ignore-signal policies reproduce uninformed values, and deterministic signal garbling cannot increase the optimized value under the registered comparison.

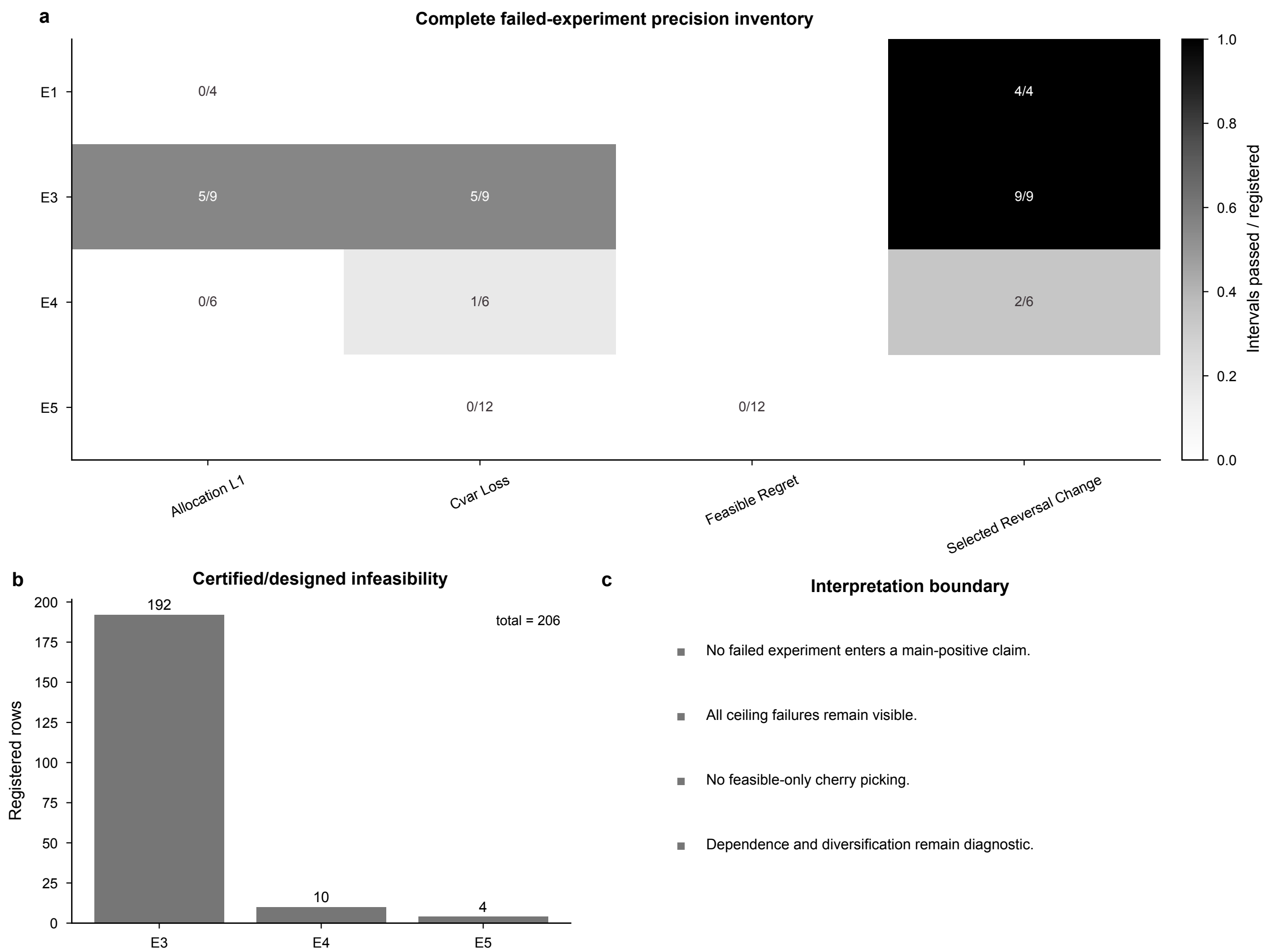

Figure 2: **Complete adverse and infeasible inventory.** Failed E1, E3, E4 and E5 estimands are shown with every 206 registered infeasible row (192 designed E3 rows and 14 certified E4/E5 rows). No feasible-only filtering enters a positive claim.

2.4 Adverse experiments and infeasibility

E1 tests margin-driven reversal, E3 the CVaR frontier, E4 dependence intensity within named families and E5 cross-law misspecification. Each has at least one registered primary interval that fails at $n = 64$, so all four remain adverse regardless of the sign or apparent precision of individual rows. The failure may arise from interval width, insufficient finite observations after registered infeasibility, or both.

There are 206 registered infeasible rows: 192 are designed E3 frontier cases and 14 are certified E4/E5 cases. Every row has a reason and remains in the ledger. No solver failure remains unexplained after excluding registered infeasibility, and no feasible-only subset is used for a positive claim.

2.5 Optimal faces, KKT checks and attribution closure

The optimal-face audit fixes the primary objective within tolerance and minimizes and maximizes each relevant acreage gap. It thereby distinguishes possible, universal and selected reversal instead of treating one solver vertex as the entire solution set. Direct loss-CVaR is recomputed from optimized scenario losses and compared with the auxiliary value. Tail weights, stationarity, complementarity and primal feasibility are evaluated independently.

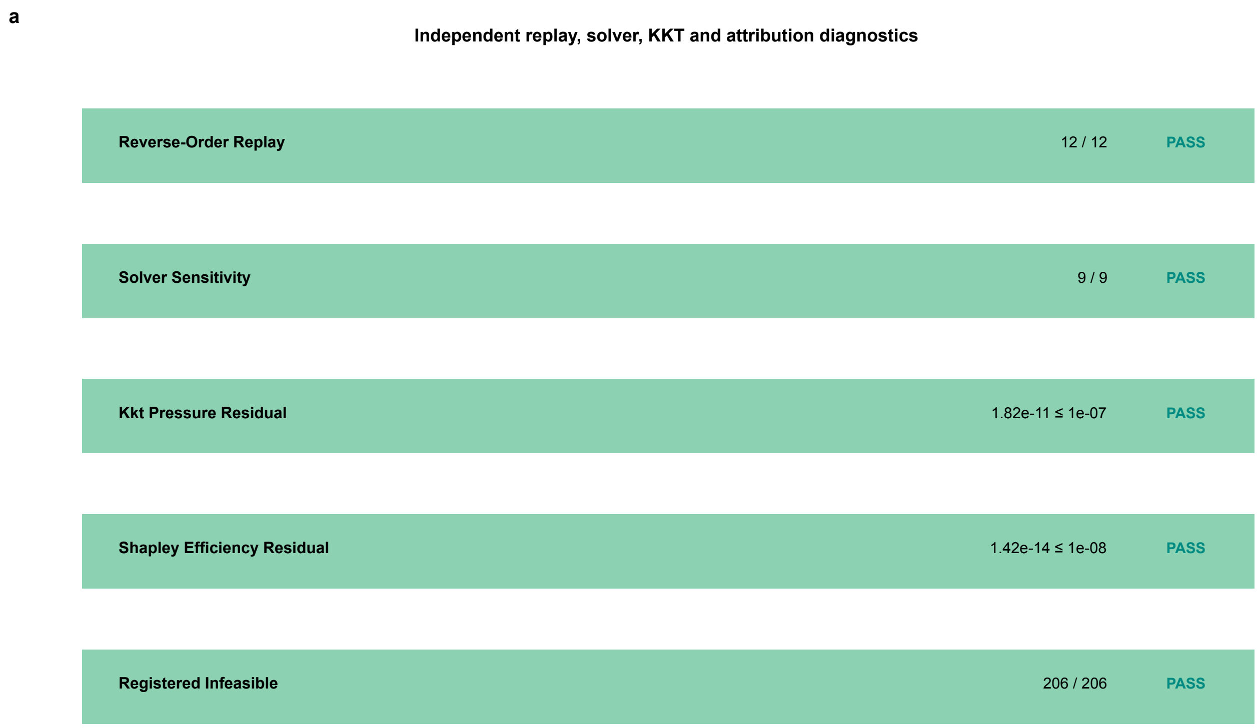

Figure 3: **Numerical diagnostics.** Reverse-order replay passes 12/12, solver sensitivity passes 9/9, the maximum KKT residual is 1.82×10^{-11}, and the maximum Shapley efficiency residual is 1.42×10^{-14}.

Reverse-order replay passes 12/12 registered checks and solver-method sensitivity passes 9/9. The maximum KKT stationarity residual is 1.82×10^{-11}. For M0–M4 attribution, all coalitions of the margins, operations, risk and dependence blocks are solved. Shapley contributions sum to the full-minus-base change with maximum efficiency residual 1.42×10^{-14}.

Sequential block paths can differ even when their endpoint is identical. Figure 4 therefore reports each block's contribution range over every registered order, with dots for order means. The all-subset Shapley value is the symmetric average of marginal contributions; it is an accounting allocation over the closed model domain, not a causal decomposition.

3 Empirical sample, estimands and robustness

3.1 Admitted panel and score definitions

The sample begins with state and US-total rows parsed from the frozen official crop summary. State rows are restricted to complete planted acreage and yield for corn, soybeans and winter wheat in a common state-year. Missing codes are not inferred. The resulting 231 crop rows form 77 state-years in 26 states. US totals are retained only for aggregation checks.

Relative yield normalizes each state crop yield by its same-year national value. Standardized revenue multiplies yield by national price. Operating- and total-cost margins subtract corresponding national cost measures after conversion to real 2024 dollars. Holding price and cost geography constant exposes state-yield variation transparently but omits local basis, marketing arrangements, heterogeneous costs and private expectations.

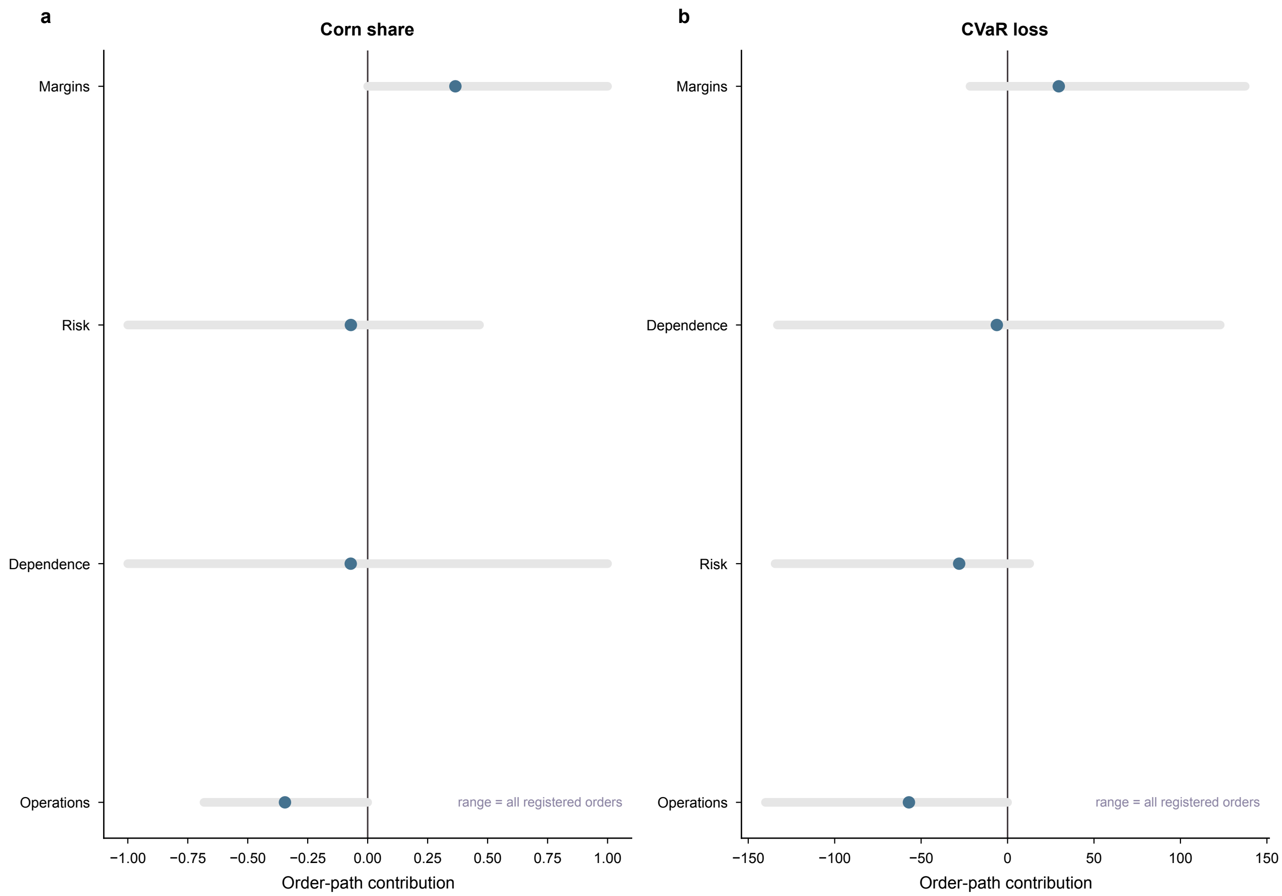

Figure 4: **Attribution order sensitivity.** Ranges cover all registered block orders and dots show order means. The display exposes path dependence rather than assigning causal meaning to one ordering.

Concurrent inversion intensity counts discordant informative pairs divided by all informative pairs. A top-rank reversal requires a unique score leader and compares it with the acreage leader; strong reversal requires the score leader to fall below every lower-scored crop in acreage. Tied score pairs are non-informative. State-cluster bootstrap intervals resample the 26 states and recompute the full estimand 5000 times.

Operating-margin inversion intensity is 0.597 [0.494, 0.697]. Across the four definitions, concurrent intensity ranges from 0.351 for standardized revenue to 0.706 for relative yield. Top-rank reversal ranges from 62.3% to 87.0%, illustrating that a reported reversal rate is conditional on the score construction.

3.2 Rolling-origin transitions

The rolling-origin design yields 51 transitions per definition. For each crop, the indicator for being the prior score leader is paired with its next-year acreage-share change. The top-minus-other contrast averages within the registered state-cluster procedure. Operating margin gives 0.0089 [-0.0018, 0.0213]; the other three intervals also include zero. Hence none of the four score definitions provides a precision-qualified positive share-response result.

The acreage leader changes in 5 of 51 transitions under every definition because the acreage ordering is common to the score definitions. Score leaders change at definition-specific rates, from 1 of 51 for standardized revenue to 23 of 51 for operating margin. The contrast is consistent with slower acreage movement than score movement, but prior acreage is only an inertia proxy. Rotation, adjustment cost, capacity and contract explanations are not observed.

3.3 State, year and aggregation checks

Leave-one-state-out estimates assess influence without asserting that states are sampled from a target superpopulation. Operating-margin inversion intensity ranges from 0.586 to 0.617 across those omissions. Year-specific estimates range from 0.547 to 0.692. Corresponding ranges for the other definitions remain separated enough to preserve definition sensitivity.

At national aggregation, operating-margin and standardized-revenue inversion intensity are zero. Total-cost margin is 0.667. Relative-yield comparisons contain 9 pairwise ties because both numerator and denominator collapse to the national yield. They are marked non-informative, not counted as agreement or reversal. Aggregation therefore changes the empirical statement and can create definitional degeneracy.

3.4 Identification ledger

The data identify descriptive score–acreage order relations, accounting constructions, their temporal ordering and their sensitivity to definition and aggregation. They do not identify observed acreage as optimal; private land, budget, rotation or contract constraints; beliefs or objective weights; a binding CVaR limit; a copula or tail-risk mechanism; causal responses; state welfare; or regret relative to a latent optimum. No such interpretation is promoted in the figures, tables or number registry.

4 Reproducibility, lineage and artifact inventory

The repository is organized as an evidence graph. Immutable raw sources are checksum-registered; processing produces canonical panels; theory, simulation and empirical modules write tidy outputs; visualization reads only verified outputs; and manuscript macros read named fields from those outputs. The path for each major statement is claim → figure or table → source data → code → configuration → execution record.

4.1 Empirical reproducibility

The Stage II empirical command starts from admitted snapshots and writes 19 canonical outputs. Two isolated executions reproduce every artifact byte for byte. The validator checks the 231 crop rows, 77 state-years, 26 states, 612 crop-transition rows, 204 rank-transition events, 5,000 bootstrap draws, score definitions, intervals, aggregation ties and the complete prohibited-inference ledger.

4.2 Simulation reproducibility

The confirmatory command resolves the frozen design hash, scenario seeds and task registry before solving. Output validators reconcile raw, contrast, face, pressure, information, attribution and stopping ledgers. Independent replay reverses task order; solver sensitivity reruns named anchors. The validator fails closed if any unexplained infeasibility, missing adverse row, promotion-rule violation, KKT excess or attribution non-closure appears.

4.3 Figure production and colour contract

The figure build exports editable SVG and PDF, 300-dpi PNG and 600-dpi TIFF for six main and four supplementary figures. Each figure has tidy source data, a caption, a manifest row and checksums. Exact dimensions are frozen. The palette is `#3D3539`, `#0F9EA8`, `#008B82`, `#45728F`, `#8CD1B2` and `#8B84A3`; lighter derived tints are allowed, and neutral grey marks adverse or unidentified evidence. Labels, ordering, markers and line styles carry the same distinctions so colour is not the only channel. Grayscale and deuteranopia proofs are generated and inspected.

4.4 Manuscript and release build

The main source is `manuscript/main.tex`; this document is `supplementary/supplementary.tex`. The manuscript-number generator records every displayed macro, unit, output file and field in a CSV registry. Claim–citation and figure-use registries restrict citations to verified literature and scientific statements to admitted evidence levels. All 44 teacher-Draft content rows retain a final disposition; unsupported historical numbers and universal mechanism claims remain excluded.

The release build compiles both documents twice in isolated directories with a fixed source date and timezone. It rejects LaTeX errors, undefined citations or references, missing assets and overfull boxes. Byte-identical PDFs are copied to the release directory, every page is rendered for visual review, font embedding is checked, and package checksums are regenerated. The final archive includes manuscript sources, Supplementary Information, six main and four supplementary figures in four formats, source data, registries, audit reports and reproduction instructions.

Software environment. Python 3.11 is managed by the locked project environment. Optimization uses SciPy/HiGHS; data processing uses pandas and NumPy; figures use Matplotlib under the frozen Stage II style contract. LaTeX uses pdfLaTeX, BibTeX and `latexmk`; Poppler renders page proofs. Exact tool versions are written to the build-environment record.

Author-owned metadata. Author names, affiliations, ORCID identifiers, funding, contributions, acknowledgements and the final competing-interests statement remain explicit placeholders. Their absence does not alter scientific results, but the package is a final scientific draft rather than a journal-submission archive until authors complete them.

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R. Tyrrell Rockafellar and Stanislav Uryasev. Conditional value-at-risk for general loss distributions. *Journal of Banking & Finance*, 26(7):1443–1471, 2002. doi: 10.1016/S0378-4266(02)00271-6. URL `https://doi.org/10.1016/S0378-4266(02)00271-6`.